CHEMISTRY

SECTION – I

Single Correct Choice Type

1. **(b)** $2A(g) \rightarrow 3B(g) + 2C(g)$

$1-\alpha \qquad \frac{3\alpha}{2} \qquad \alpha$

$$\text{Slope} = \frac{n_T R}{V} = \frac{\left(1-\alpha+\frac{3\alpha}{2}+\alpha\right)\times 0.08}{0.16} = \frac{\left(1+\frac{3}{2}\times 0.4\right)}{2} = 0.8$$

2. **(b)** This is because H and Br are trans in position in compound (B). Among (A) and (C), (A) is a vinyl bromide while (C) is a vinyl chloride, so bromide undergoes reaction faster than chloride. (D) reacts slowly because the H is least acidic in it.

3. **(d)** $\frac{W_{acid}}{M_{acid}} = 50\times 0.4\times 10^{-3}$

$$\therefore \quad M_{acid} = \frac{W_{acid}}{50\times 0.4\times 10^{-3}} = \frac{4\times 1000}{50\times 0.4} = 200 \text{ g mol}^{-1}$$

$$\therefore \text{ Mol of acid} = \frac{4}{200} = \frac{1}{50} = 2\times 10^{-2}$$

Mole of base used in titration = $0.40 \times 25 \times 10^{-3} = 10^{-2}$

$\therefore$ Mole of salt formed = 1×10^{-2}

Mole of acid left = 1×10^{-2}

$$\therefore \quad pH = pK_a + \log\frac{S}{A} = pK_a = 4 \quad \left(pH = pK_a + \log\frac{S}{A}\right)$$

4. **(d)** $\underset{(X)}{BeO} + C + Cl_2 \xrightarrow{1000\ K} \underset{(Y)}{BeCl_2} + CO$

$\underset{(Y)}{BeCl_2} + 2H_2O \rightarrow Be(OH)_2 + 2HCl$

5. **(b)** $\kappa_{soln} = \kappa_{CaF_2} + \kappa_{water}$

$\kappa_{CaF_2} = \kappa_{soln} - \kappa_{water} = 4.2 \times 10^{-5} - 0.2 \times 10^{-5} = 4 \times 10^{-5}$ S cm^{-1}

$\underset{a-s}{CaF_2(s)} \rightleftharpoons \underset{s}{Ca^{2+}(aq)} + \underset{2s}{2F^-(aq)}$

$K_{sp} = 4s^3$, $[F^-] = 2[Ca^{2+}]$

$\kappa_{CaF_2} = \kappa_{Ca^{2+}} + \kappa_{F^-}$

$$\kappa_{CaF_2} = \frac{\Lambda_{m,Ca^{2+}}[Ca^{2+}]}{1000} + \frac{\Lambda_{m,F^-}[F^-]}{1000}$$

$$\kappa_{CaF_2} = \frac{\Lambda_{m,Ca^{2+}}[Ca^{2+}]}{1000} + \frac{\Lambda_{m,F^-}2[Ca^{2+}]}{1000} = \frac{[Ca^{2+}]}{1000}\left(\Lambda_{m,Ca^{2+}} + 2\Lambda_{m,F^-}\right)$$

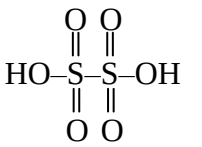
$4 \times 10^{-5} \times 1000 = [Ca^{2+}]\ \ (\Lambda^{\infty}_{m,\,Ca^{2+}} + 2\Lambda^{\infty}_{m,\,F^-}) = [Ca^{2+}]\,(104 + 2 \times 48)$

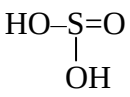
$4 \times 10^{-2} = [Ca^{2+}]\,(200)$

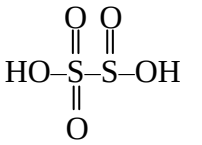
$[Ca^{2+}] = s = \dfrac{4 \times 10^{-2}}{200} = 2 \times 10^{-4}\ M$

$\therefore\ K_{sp} = 4s^3 = 4(2 \times 10^{-4})^3 = 32 \times 10^{-12} = \mathbf{3.2 \times 10^{-11}\ M^3}$.

6. **(b**

$H_2S_2O_6$  4π bonds

H_2SO_3  1π bond

$H_2S_2O_5$  3π bond

7. **(b)** From the unit of Henry's law constant for N_2 gas, it implies that

$$X_{N_2} = k_{N_2} p_{N_2}$$

$$X_{N_2} = 1 \times 10^{-5} \times 5 \times \frac{80}{100} = 4 \times 10^{-5}$$

$$\therefore\ \frac{n_{N_2}}{n_{N_2} + n_{H_2O}} = 4 \times 10^{-5}$$

Since the mole fraction of dissolved N_2 is very small, so $n_{N_2} + n_{H_2O} \simeq n_{H_2O}$

$$\therefore\ \frac{n_{N_2}}{n_{H_2O}} = 4 \times 10^{-5}$$

$n_{N_2} = 4 \times 10^{-5} \times 10 = \mathbf{4 \times 10^{-4}\ moles}$.

8. **(a)** $P_4O_{10} + 6\ H_2O \rightarrow 4H_3PO_4$

Mass of water added is 27 g.

Moles of H_2O added = $\dfrac{27}{18} = \dfrac{3}{2}$

Moles of P_4O_{10} that react with $\dfrac{3}{2}$ moles of H_2O = $\dfrac{3}{2} \times \dfrac{1}{6} = \dfrac{1}{4}$

$\therefore$ Mass of P_4O_{10} in the sample = $\dfrac{1}{4} \times 284 = \mathbf{71\ g}$.

SECTION – II
Multiple Correct Choice Type

9. (a, b, d)

Aldehydic linkage is oxidized by Br_2 water. Hemiacetal dissociate in aqueous solution to give free –CHO group.

10. (a, b, c, d)

11. (a, c, d) $[A] + [B] + [C] = 1$ M

$$\therefore \quad K_3[C] + \frac{[C]}{K_2} + [C] = 1$$

$$\text{or,} \quad [C]\left\{0.6 + \frac{1}{0.4} + 1\right\} = 1$$

$$\text{or,} \quad [C] = \frac{1}{4.1}\text{M}$$

$$\text{Besides, } [C] = K_2[B] = \frac{[A]}{K_3}$$

$$\therefore \quad K_2 \times K_3 = \frac{[A]}{[B]}$$

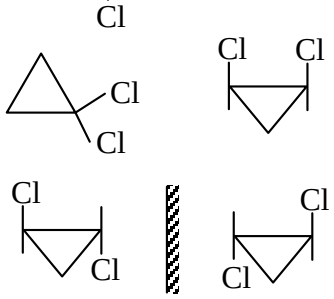
$$\text{or} \quad 0.4 \times 0.6 = \frac{1}{K_1} \quad \Rightarrow \quad K_1 = \frac{1}{0.24}$$

12. (a, b, c) A has 11 isomers including stereoisomers.

$CH_3–CH=CCl_2$ $ClCH_2–CH=CHCl$ (geometrical isomerism)

$Cl_2CH–CH=CH_2$ $ClCH_2–\underset{\displaystyle Cl}{\underset{|}{CH}}=CH_2$

$CH_3–\underset{\displaystyle Cl}{\underset{|}{C}}=CHCl$ (geometrical isomerism)



All isomers are reducible, give addition products and can undergo free radical substitution reaction.

SECTION–III
Comprehension Type

Passage–I

$$2[Ag(CN)_2]^- + Zn \longrightarrow 2Ag + [Zn(CN)_4]^{2-} \; ; \; E_1^o = -0.31 - (-1.26) = 0.95\text{V}$$

$$2[Au(CN)_2]^- + Zn \longrightarrow 2Au + [Zn(CN)_4]^{2-} \; ; \; E_2^o = -0.60 - (-1.26) = 0.66\text{V}$$

$E_1^o > E_2^o$, thus Zn reduces $[Ag(CN)_2]^-$ in preference to $[Au(CN)_2]^-$.

13. (c) Moles of $[Ag(CN)_2]^-$ in 500 L solution = 500 × 0.003 = 1.5

Moles of $[Au(CN)_2]^-$ in 500 L solution = 500 × 0.01 = 5

Moles of Zn added = $\frac{40}{65.4} = 0.61$

According to reaction, 0.61 mole of Zn will consume 1.2 mole of $[Ag(CN)_2]^-$, thus moles of $[Ag(CN)_2]^-$ left = 1.5 – 1.2 = 0.3.

Conc. of $[Ag(CN)_2]^-$ after the reaction = $\frac{0.3}{500/3} = 0.0018$ M.

Conc. of $[Au(CN)_2]^-$ after the reaction = 0.01 × 3 = 0.03 M.

14. (b) $Au^+(aq) + 2CN^-(aq) \rightleftharpoons [Au(CN)_2]^-(aq)$

$$K_f = \frac{[Au(CN)_2]^-}{[Au^+][CN^-]^2} = 4 \times 10^{28}$$

$$\frac{[Au(CN)_2]^-}{[Au^+]+[Au(CN)_2]^-} \times 100 = 99,\ [Au(CN)_2]^- = 99[Au^+]$$

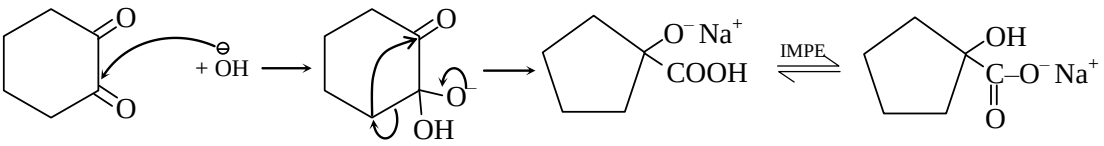
$$\therefore\ K_f = \frac{99[Au^+]}{[Au^+][CN^-]^2}\ ;\ \frac{99}{[CN^-]^2} = 4 \times 10^{28}$$

$$[CN^-] = \sqrt{\frac{99}{4\times10^{28}}} = 4.97 \times 10^{-14} \simeq \mathbf{5 \times 10^{-14}\ M}.$$

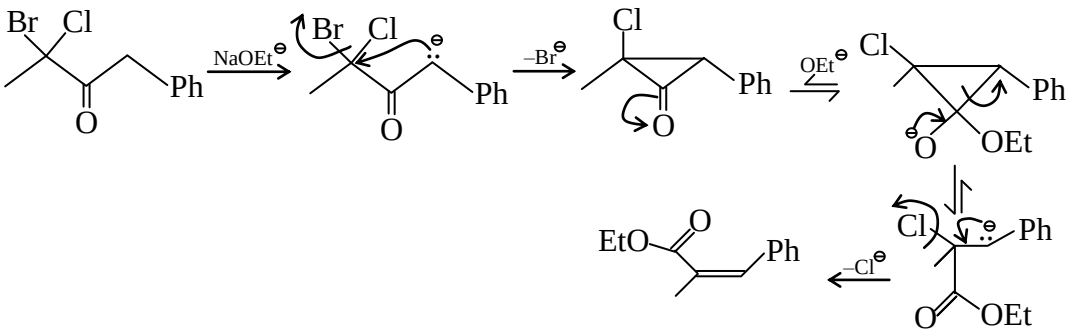
15. (d) This process of gold extraction is not suitable as CN^- passes into the ground water and releases toxic HCN in it, making the human life at high level of risk.

Passage–II

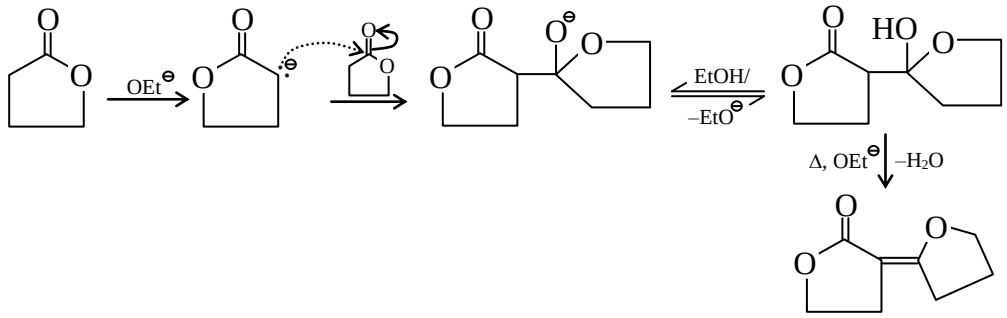
16. (a)



17. (c)



18. (d)



SECTION–IV
Matrix–Match Type

1. **(A) – 3, 4** ; $SbCl_3 + H_2O \longrightarrow \underset{\text{orange ppt.}}{SbO^+Cl^-\downarrow} + 2HCl$
(colourless)

The reaction involves hydrolysis and colour change.

(B) – 1, 4 ; $2H^+ + 2CrO_4^{2-} \rightleftharpoons Cr_2O_7^{2-} + H_2O$
(yellow) (orange)

The equilibrium can be shifted in forward direction by decrease of pH (i.e. on acidification).

(C) – 2, 4, 5 ; $2KMnO_4 + 2KOH \longrightarrow 2K_2MnO_4 + H_2O + O$
(purple) (green)

The reaction occurs by addition of a base, it involves reduction of Mn from +7 to +6 oxidation state and colour changes from purple to green.

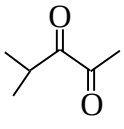
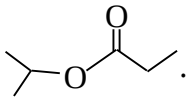
(D) – 1, 3 ; $Na_2B_4O_7 + 2HCl + 5H_2O \longrightarrow 2NaCl + 4H_3BO_3$

This reaction is occuring in presence of a mineral acid and it is a hydrolysis reaction.

2. **(A) – 3** ; Imines can be reduced by $NaBH_4$ readily.

(B) – 1, 2, 3, 5 ; SeO_2 will oxidise it into 1,2-diketone, Zn-Hg/HCl will selectively reduce keto functionality to CH_2, $NaBH_4$ will selectively reduce $\rangle C{=}O$ to $\rangle CH{-}OH$ grouping. CF_3CO_3H will oxidise it via Baeyer-Villiger oxidation.

(C) – 1, 4 ; SeO_2 oxidises alkene at allylic position and introduce OH on allylic carbon. Hydrogenation by Wilkinson's catalyst can reduce double or triple bonds selectively without affecting $-NO_2$, $\rangle C{=}O$, $-CO_2H$ groups etc.

(D) – 1, 2, 3, 5 ; SeO_2 oxidises it to  . Clemenson and $NaBH_4$ reduction for this compound is also feasible. CF_3CO_3H will cause Baeyer-Villiger oxidation to give  .